

Analyzing and Visualizing Country-Level Waste Data

Problem Statement & Background

Solid waste management is a critical issue that affects every single person on the planet. Whether individuals are responsible for managing their own waste or governments are tasked with implementing large-scale waste management systems, the implications are universal. With increasing urbanization, economic development, and population growth, the challenge is only expected to intensify. The World Bank projects that global waste generation will surge from 2.01 billion tons in 2016 to 3.40 billion tons in 2050. Alarmingly, over 33% of this waste is currently mismanaged, ending up in open dumps or being burned, leading to severe environmental and health consequences.

This project aims to analyze country-level data on GDP (Gross Domestic Product), population, waste composition, and total MSW (Municipal Solid Waste generated) to identify correlations, specifically what impacts national-scale waste production, that can guide better waste management policies.

Introduction

I've always been passionate about understanding and addressing the environmental challenges that arise from human activity. Climate change, one of the most pressing issues of our time, is largely driven by anthropogenic factors. Among these, waste mismanagement is a critical but often overlooked contributor. Improper disposal of waste not only pollutes ecosystems but also generates significant greenhouse gas emissions, exacerbating global warming.

This project represents a step towards tackling this issue. By analyzing waste generation patterns across countries, we can better understand the links between economic development, population growth, and waste production.

Description of Data

This project uses data from the What a Waste Project Global Database, created by the World Bank Group. It's a rich resource for understanding global waste patterns and management practices. The dataset brings together waste-related information for over 200 countries, covering everything from how much waste is generated to its composition.

The dataset consists of **217 rows** (countries and territories) and **10 columns**, including:

Column Number	Column Name in CSV file	Data description
1	country_name	List of all countries and territories analyzed
2	gdp	Gross Domestic Product of each country in U.S. dollars
3	composition_food_organic_waste_percent	Breakdown of waste into six categories - Organic Waste, Glass, Metal, Paper, Plastic, Other Waste
4	composition_glass_percent	
5	composition_metal_percent	
6	composition_other_percent	
7	composition_paper_cardboard_percent	
8	composition_plastic_percent	
9	population population number of people	Total population of country
10	total_msw_total_msw_generated_tons_year	Annual amount of waste generated by each country, measured in tons.

Some rows had gaps, which were not overlooked—for analysis, these missing values were handled appropriately, which is further discussed in the methods.

Methods

In this project, several steps were taken to clean, analyze, and visualize country-level waste data in comparison to GDP and population, in a way that makes it both informative and engaging for the user.

Data Cleaning and Standardization: The first task was to prepare the data. This involved loading the dataset from the CSV file containing all the data into a structure called a DataFrame, which makes it

easier to manipulate and analyze. The column names were standardized into simple positional indices for clarity. Any rows with missing country names were also discarded at this point, ensuring that the dataset was free of incomplete entries.

Extracting Country-Specific Data: The user was asked to input the name of a country of their choice. This user-specified country's data was extracted - GDP, population, total waste generated (MSW), and the breakdown of waste by category (such as organic waste, plastic, metal, glass, paper, and other). The method worked by filtering the dataset for a specific country and pulling the relevant figures. If the country didn't have enough data (i.e., missing key values), the function would return an error message for the user, ensuring that the analysis wasn't performed on incomplete data.

Visualizing Waste Composition with Pie Charts: After extracting the necessary data, the waste composition generated by the user-specified country was visualized through pie charts.

Per Capita Waste Calculation: To understand how much waste an individual in a country is generating on average, the project calculated per capita waste. This was done by dividing the total waste generation by the population of the country. This is used to compare to user's data which is later collected.

User Waste Data Collection and Comparison: One of the more interactive parts of the project was the ability to compare user-generated waste with national averages. The user was prompted to input their own waste generation data, including the total amount of waste they produce each year and the percentage that they recycle. Then their personal waste contribution compared to the national per capita waste average was calculated. This helps the user reflect on their personal impact on waste generation and spark thoughts on potential improvements.

Ranking Countries by Waste and GDP: Countries were ranked from the lowest to the highest in terms of both their economic output (GDP) and total waste generated. This allowed users to see how their country measured up in global rankings, helping to identify trends where wealthier countries may be generating more waste.

Scatter Plot Analysis: The relationships between GDP, population, and waste generation of all countries were visualized through scatter plots, with the country selected by the user highlighted for comparison. By doing this, trends between the variable were illustrated.

Correlation Analysis: Statistical correlations between GDP, population, and waste generation indicated how strongly these variables were related to each other. For example, it helped assess whether higher GDP led to more waste generation, or whether a larger population directly translated to increased waste. The correlation matrix generated during this step was also visually displayed.

Results & Conclusion

These results are based on data from countries as listed in the CSV file, as mentioned before. However for the graphs based on the user-specified country, the visualizations below have been based on the input being ‘Greece’.

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Enter country name: greece
Greece GDP world rank (lowest to highest): 152
Greece Waste Generation rank (lowest to highest): 166
Approx. how much waste does your household generate every year? (tons)
10
Approx. what percentage of the waste do you recycle?
5
You have an above-average waste generation! RECYCLE!
You contribute 0.00017 % of the Municipal Solid Waste generated by Greece
Correlation between GDP and MSW: 0.041312873873372666
Correlation between GDP and Population: -0.061642058758295495
Correlation between Population and MSW: 0.8796146115767257
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Figure 1 Output after full program execution

The analysis given in Fig 1, complements the visualizations that will follow by providing quantitative insights into Greece’s waste management status. Greece ranks 152nd in GDP globally but 166th in waste generation, indicating that the country produces a disproportionately high volume of waste relative to its economic standing.

A sample user enters that their annual household waste production is 10 tons, with a recycling percentage of 5% of their waste. They demonstrate above-average waste generation in Greece that urges them to improve their personal waste management habits by recycling. The solution to such a grand-scale problem begins with the individual. Thus this interactive section of the program was

necessary to allow an individual to understand their own impact on this issue and invoke a sense of change in their habits.

Then, correlation values from data of all countries emphasizes that population size strongly drives waste generation (0.879 correlation between population and MSW), while GDP has little influence on waste production (0.041 correlation between GDP and MSW). There is also a weak association between GDP and population (-0.061), dispelling the assumption that more economically developed countries inherently produce more waste due to larger populations.

The analysis concludes that Greece should enhance its recycling systems and adopt waste reduction policies to address its significant contribution to global waste generation. Similarly, this could be applied to all other countries as well depending on a user's input country.

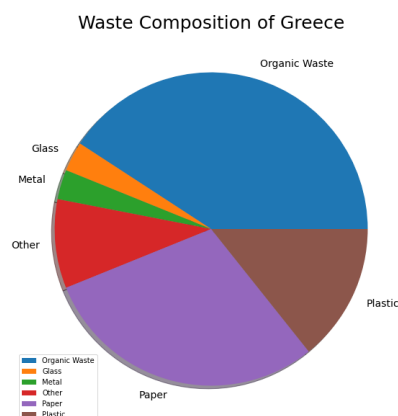


Figure 2 Waste Composition of User-specified country - Greece

The pie chart as seen in Fig 2 provides a breakdown of the types of waste generated in Greece, for a clear visual representation of the composition of the country's municipal solid waste. Organic waste constitutes the largest proportion, dominating the chart, followed by plastic and paper, which together form significant portions of the waste stream. Smaller categories, such as glass, metal, and other waste, make up the remainder.

Thus, Greece should consider implementing more composting or anaerobic digestion practices, which could reduce the volume of waste sent to landfills while also generating renewable energy or compost for agricultural use. By understanding the waste composition, stakeholders and

policy makers can design and implement waste reduction policies and technologies tailored to the specific challenges and opportunities in Greece.

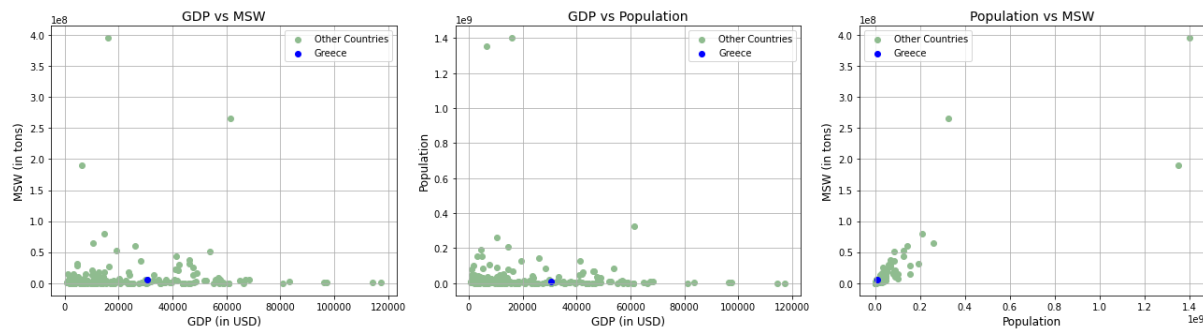


Figure 3 Scatterplots containing data points of all countries. (Left to Right) GDP vs MSW, GDP vs Population, Population vs MSW

The first scatterplot (Fig 3) depicts the relationship between GDP (measured in USD) and total MSW generated (in tons). Countries with varying GDP levels show similar amounts of waste generation, indicating that economic output has minimal influence on MSW. Greece, marked in blue, falls within the general cluster of other countries, showing a similar extent of deviation of waste generation from its GDP.

The second scatterplot (Fig 3) depicts the relationship between GDP and Population. Here too, the data reveals no strong correlation. Some countries with low populations exhibit high GDP levels, while others with larger populations show moderate or low GDP values. Greece, again represented by the blue marker, fits comfortably with the other nations.

The third scatterplot (Fig 3) however, depicting the relationship between Population and MSW generation, shows a clear positive trend: as population size increases, the amount of MSW generated also rises significantly. Greece is positioned among countries with smaller populations and moderate levels of waste generation. Thus, population size is concluded to be a primary driver of waste production in countries.

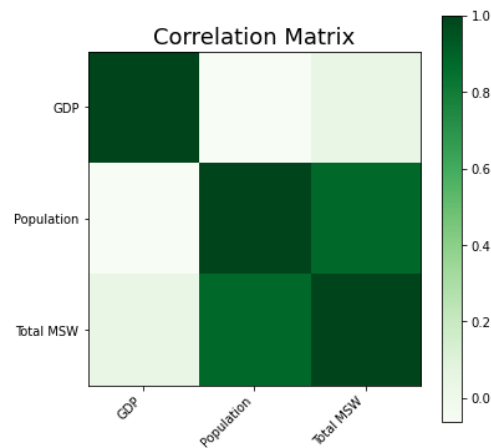


Figure 4 Correlation Matrix depicting correlation values between all three variables (GDP, MSW & Population)

The correlation matrix as seen in Fig 4 uses a heat map where the intensity of the color signifies the strength of the correlation between two variables. The correlation between GDP and Total MSW is weakly positive (0.041), indicating that economic output has no significant influence on waste generation. The relationship between Population and Total MSW, however, is strongly positive (0.879), showing that as population increases, the volume of waste generated also rises. The weak negative correlation (-0.061) between GDP and Population shows that a country's population size does not affect its GDP.

Thus, the focus toward addressing the waste impacts of population growth over GDP increases. This sets a foundation for governments and policy makers to create waste management strategies that prioritize high-population areas.

Strengths

The use of scatterplots and a correlation matrix made the analysis visually intuitive and easy to interpret. By including data from multiple countries, the project adopted a global perspective on waste management.

As for personal strengths while working on the project, Whenever I ran into an issue—like a graph not working or the data not matching up—I didn't get stuck for long. I stayed focused, tried

different approaches, and found solutions. This helped me keep the project moving forward, even when things didn't go as planned.

Limitations

One major limitation of the analysis was missing values in data. A more complete dataset would enhance the robustness of the correlations and allow for more accurate interpretations of the relationships between variables.

The dataset was static, representing a single snapshot in time. As a result, the analysis could not capture trends in waste generation, economic growth, or population dynamics over time. A time-series dataset would allow for a deeper analysis and also allow forecasting amounts of future waste production if trends persist.

As for personal limitations, there were some tools or methods used in this project that were new to me. I had to spend time learning how to use them, which sometimes slowed things down. For example, creating certain types of graphs or fixing issues with the data took more effort than I initially expected.

Future Work

Adding historical data to the dataset would make it possible to study how waste generation, population, and economic factors have changed over time. This would help identify patterns, like whether waste generation increases faster in growing economies or slows down in developed countries. With this information, we could also make predictions about future trends in waste management.

Bringing in more data, like recycling rates, government waste policies, or waste management infrastructure, could give a better understanding of how different countries handle waste. For example, we could see how stricter recycling rules or better infrastructure reduce waste and apply those lessons to other places.

Citations

The World Bank. “What a Waste.” *Datatopics.worldbank.org*, 2018, datatopics.worldbank.org/what-a-waste/.

Blagoeva, Nadezhda, et al. “Relationship between GDP and Municipal Waste: Regional Disparities and Implication for Waste Management Policies.” *Sustainability*, vol. 15, no. 21, 1 Jan. 2023, p. 15193, www.mdpi.com/2071-1050/15/21/15193, <https://doi.org/10.3390/su152115193>.